# CloudboyX Enterprise Platform

## Complete Architectural Module Breakdown & Operations Manual

Version 4.2.0 · Comprehensive Enterprise Edition

**System Architecture Summary**

The CloudboyX Platform is a unified, autonomous enterprise orchestration suite engineered for cross-channel retail, quantitative prediction, minimum advertised price (MAP) brand defense, double-entry financial ledger accounting, and visual website publishing. This document details each constituent module, technical architecture, and operational how-to instructions.

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*Document generated automatically from live codebase specifications and integration contracts.*

CORE LEDGER MODULE

# Module 01: Financial Ledger & GAAP Double-Entry Engine

*Immutable balance sheet verification, automated journal entries, and real-time trial balance balancing.*

## Overview & Architectural Purpose

The Financial Ledger module serves as the immutable system of record. Every inbound transaction is converted into standardized balanced double-entry journal entries adhering to GAAP/IFRS principles. Total debits must always equal total credits before any transaction can commit to the relational database.

## Key Capabilities

- **Double-Entry Integrity:** Strict mathematical balance verification ensures Debits = Credits for all entries.
- **In-Memory Relational SQL:** Powered by AlaSQL for near-instant balance aggregation across Accounts and Entries.
- **Multi-Format Financial Exports:** Export full general ledgers into structured Vector PDF, CSV tables, or QuickBooks IIF format.
- **Interactive Adjustment Forms:** Post manual debit/credit corrections with dedicated entity tagging and audit logs.

## How-To Instructions: Step-by-Step

### 1. Access the Financial Ledger

Click "01. Ledger Gateway" in the anchored top menu or select "Strategic Ledger Health" from the Central Access Hub.

### 2. Review Real-Time Balance Indicators

Examine the Strategic Ledger Health cards showing Total Debits, Total Credits, Gross Invoiced Volume, and Active Entities.

### 3. Execute Transaction Reversals or Adjustments

Scroll to the Journal Entries table. Click "Reverse Entry" on any posted transaction to generate an offsetting reversing journal entry.

### 4. Export Reports

Click the "Export Vector PDF", "Export CSV", or "Export QuickBooks (.IIF)" buttons in the Ledger toolbar for external financial reporting.

INTEGRATION GATEWAY

# Module 02: Universal Ingress Gateway & Optical QR / Barcode Scanner

*Multi-channel feed transformation, optical camera parsing, and canonical schema mapping.*

## Overview & Ingestion Pipeline

Translates raw order payloads from disparate systems—Shopify JSON, EDI ANSI X12 850 documents, legacy CSV feeds, and optical QR/barcode scans—into an immutable CanonicalOrder schema with integer cent-precision line items.

## Optical Camera Scanner Features

- **Real-Time WebRTC Video Stream:** High-speed camera viewfinder with automatic optical target bounding brackets.
- **Torch & Hardware Zoom:** Integrated flashlight toggle and pinch/slider zoom control for scanning physical warehouse labels.
- **Thermal & Grayscale Filters:** Real-time Canvas image processing filters to enhance faded or low-contrast barcodes.
- **Recent Scans History & Sorting:** Persistent scan history with sorting by Date (Newest/Oldest) and Favorite pattern pinning.
- **Scanning Instructions Help Modal:** Comprehensive user guide with tips for lighting, angle, and distance.

## How-To Instructions: Ingesting Orders

### 1. Launch Scanner or Ingress View

Click "02. Connect Gateway" or click the "Scan QR Code" camera button.

### 2. Select Data Channel Feed

Choose Shopify, EDI 850, Legacy CSV, or Optical Scanner from the feed tabs.

### 3. Execute Optical Scan

Align any order QR code inside the camera viewfinder. The optical engine automatically parses, validates, and stores the order.

### 4. Pin Favorite Patterns & Sort

Click the Star icon to mark repetitive vendor patterns as favorites. Use the "Sort" dropdown to view Newest First, Oldest First, or Favorites First.

BRAND DEFENSE

# Module 03: Autonomous MAP & Brand Defense Intelligence Suite

*24/7 retail reseller pricing enforcement, automated threat grading, and Cease & Desist dispatching.*

## Overview & Threat Enforcement

Protects brand integrity and authorized distributor margins by monitoring Minimum Advertised Price (MAP) thresholds across Amazon, Walmart, eBay, Target, and unauthorized third-party gray market storefronts.

## Key Capabilities

- **Reseller Network Monitoring:** Tracks 16+ authorized and rogue seller accounts with risk scores and compliance ratings.
- **Dynamic MAP Audit Engine:** Compares advertised prices against MSRP rules and computes exact revenue leakage.
- **Automated Cease & Desist Pipeline:** Generates tiered legal enforcement notices: Tier 1 Warning, Tier 2 Formal C&D, Tier 3 Supply Suspension.
- **Live Marketplace Crawler Simulator:** Test price drops and instantly observe system alerts and automated compliance triggers.

## How-To Instructions

### 1. Navigate to MAP Defense

Select "04. MAP Defense" from the top menu or the Central Access Hub.

### 2. Inspect High-Risk Resellers

Sort the Reseller table by "Risk Level: Critical". Review the flagged price deviations below approved MAP minimums.

### 3. Generate Legal Enforcement Notice

Click "Issue C&D Notice" on the flagged violator. Select the appropriate tier and dispatch the formal enforcement letter.

### 4. Run Real-Time Compliance Crawl

Click "Run Market Price Audit" to scrape and recalculate compliance scores across all protected SKUs.

QUANTITATIVE ENGINE

# Module 04: CloudboyX Predict & Bayesian Odds Analytics Engine

*Quantitative event forecasting, AI bull/bear debate consensus, and Kelly criterion expected value (EV).*

## Overview & Forecasting Algorithms

The Predict Engine applies Bayesian statistical inference and dual-agent AI debate (Bull Agent vs. Bear Agent) to calculate true market probabilities, expected values (+EV), and mathematically optimal Kelly bet sizing across supply chain events, market contracts, and macro trends.

## Core Capabilities

• **Dual AI Debate Matrix:** Bull and Bear LLM reasoning agents challenge each other with historical data to establish probability bounds.

• **Kelly Criterion Calculator:** Computes full, half, and fractional Kelly bankroll allocations to maximize geometric growth.

• **Interactive Probability Sliders:** Adjust underlying win/loss probabilities in real time to observe dynamic Expected Value curves.

• **Contract Ledger History:** Maintains an auditable log of predictive contracts, settlement outcomes, and realized yield.

## How-To Instructions

### 1. Open Predict Engine

Click "05. Predict" in the header navigation.

### 2. Select or Create an Event Contract

Choose an active prediction market or supply contract from the list.

### 3. Trigger AI Bull/Bear Debate

Click "Run Bayesian Debate". Review the opposing analytical arguments and the synthesized probability score.

### 4. Calculate Optimal Sizing

Input your total allocated bankroll. The system outputs exact recommended position sizes using the fractional Kelly formula.

VISUAL PAGE BUILDER

# Module 05: CloudboyX Design Studio (Bootstrap 5 Site Builder)

*Visual drag-and-drop website creation, 108+ pre-built templates, dynamic Google Fonts, and HTML export.*

## Overview & Studio Architecture

CloudboyX Design Studio is a complete visual web design environment built with Bootstrap 5.3 semantics. It features fluid 100% full-screen canvas editing, simulated mobile/tablet device frames, a dynamic Google Fonts picker, an AI Command Palette (Ctrl+K), agency white-label branding, and zero-vendor-lock-in HTML5 export.

## Key Studio Capabilities

- **Fluid Responsive Viewports:** Toggle between Desktop (Fluid 100%), Tablet (768px frame), and Mobile (390px frame).
- **108+ Pre-Built Templates:** One-click starter layouts for SaaS, E-Commerce, Portfolios, Consulting, and Enterprise Landing Pages.
- **AI Semantic Block Search:** Press Ctrl+K or # K to describe any section in plain English and have Gemini Flash recommend matching blocks.
- **Google Fonts Specimen Picker:** Search 1,500+ Google Fonts with live specimen previews for heading and body typography.
- **Agency White-Label & Client Portal:** Customize agency branding, invite client reviewers, and issue recurring invoices.
- **Clean Semantic HTML5 Export:** Export standard, self-contained Bootstrap 5.3 HTML5 + CSS3 with zero proprietary dependencies.

## How-To Instructions

### 1. Launch Design Studio

Click "06. Design Studio" or the dedicated "Builder" button in the top navigation.

### 2. Select a Template or Start Blank

Click "Template Gallery" to preview and clone any of the 108+ pre-built industry layouts.

### 3. Insert & Customize Blocks

Hover between sections and click "+ Quick Insert" to add Heros, Features, Pricing, Testimonials, or Contact Forms.

### 4. Test Across Viewports

Click the "Desktop", "Tablet", and "Mobile" buttons in the top center bar to verify responsive layout cascades.

### 5. Export HTML / Exit

Click "Export HTML" in the studio header to download the clean web bundle, or click "Exit to Main Suite" to return.

SUPPLY CHAIN PORTAL

# Module 06: Procurement Portal & Supply Chain Management

*Purchase order lifecycles, vendor catalog cross-referencing, multi-currency conversion, and inventory fulfillment.*

## Overview & Purchase Order Lifecycle

The Procurement Portal manages vendor master records, line-item purchase orders (POs), multi-stage approval workflows, and warehouse fulfillment tracking with automatic posting into the General Ledger accounts payable upon receipt.

## How-To Instructions

### 1. Open Procurement Portal

Navigate to "Financial Ledger" dropdown and select "Procurement Portal".

### 2. Create a Purchase Order

Click "+ New Purchase Order". Select the approved vendor, currency (USD, EUR, GBP), and specify line-item SKUs, quantities, and agreed unit costs.

### 3. Submit for Approval

Change PO status from "Draft" to "Submitted". Authorized administrators can then approve the PO.

### 4. Receive Goods & Reconcile

Upon warehouse receipt, mark the PO as "Received" to automatically generate the accounts payable voucher in the general ledger.

ENTERPRISE GAAP

# Module 07: Multi-Entity GAAP Accounting & Intercompany Elimination

*Parent/subsidiary corporate structures, currency normalization, and intercompany ledger reconciliation.*

## Overview & Consolidation Rules

Enables enterprise organizations with international subsidiaries (e.g. US Domestic, EU Logistics BV, UK Trading Ltd) to roll up disparate regional ledgers into a single consolidated financial statement while automatically eliminating offsetting intercompany loans and management fees.

## How-To Instructions

### 1. Access Multi-Entity View

Select "Multi-Entity GAAP" from the Financial Ledger category in MegaNavbar.

### 2. Select Active Entity Filter

Filter the general ledger by specific corporate entity or select "Consolidated Rollup (All Entities)".

### 3. Execute Intercompany Elimination

Click "Run Intercompany Elimination" to generate GAAP elimination splits that cancel out intercompany balances before financial reporting.

SYSTEM OPERATIONS

# Module 08: Diagnostics, SQL Console Workbench & Audit Trail

*Direct SQL relational query workbench, memory ledger diagnostics, and real-time security event telemetry.*

## Overview & Diagnostic Tools

Provides systems administrators and financial auditors direct diagnostic access to the in-memory relational SQL tables (accounts, transactions, journal_entries) with interactive SQL query execution, pre-built GAAP verification queries, and immutable security audit logs.

## How-To Instructions

### 1. Open SQL Workbench

Click "Diagnostics & Admin" in MegaNavbar or select "SQL Console Workbench".

### 2. Run Pre-Built Verification Queries

Select from the Quick Query dropdown (e.g., "Verify Debit/Credit Balance Equality", "Unbalanced Transactions Check", or "Account Activity Breakdown").

### 3. Execute Custom SQL Queries

Type standard SQL in the editor (e.g., SELECT * FROM accounts WHERE balance > 0) and click "Execute SQL Query".

### 4. Review Audit Trail

Inspect the live real-time security audit trail timeline for administrative actions, permission changes, and transaction postings.

SECURITY & ACCESS

# Module 09: Authentication, Granular RBAC & Sandbox vs. Production

*Firebase security rules, role permissions, and environment sandboxing with persistent test seeds.*

## Overview & Security Model

Secures platform operations with Firebase Authentication and client-side role-based access control (RBAC). Provides a persistent "Test Sandbox" environment so demo data, mock orders, and test vendors persist across GitHub export and external server deployments.

## How-To Instructions

### 1. Sign In or Switch Roles

Click the User Profile badge in the top right to open the Auth & Account modal. Sign in via Google or Email.

### 2. Toggle Environment (Sandbox vs Production)

Click the "Ø>Ýê Test Sandbox" badge in the navigation bar to open the Environment Switcher. Switch to Live Production or return to Test Sandbox with preserved test seed data.

### 3. Configure Reseller Co-Branding Lock

In Master Admin Mode, click the Lock icon next to footer fields to prevent downstream resellers from tampering with licensing or compliance notices.

PLATFORM NAVIGATION

# Module 10: Central Access Menu Hub & Public Marketing Website

*Unified executive launchpad, module permission discovery, and public product marketing portal.*

## Overview & Navigation Hierarchy

Provides a clean, intuitive executive dashboard that organizes all ten specialized modules into clear operational cards. Also includes the standalone public marketing landing page accessible to non-authenticated visitors.

## How-To Instructions

### 1. Open Central Access Hub

Click "Access Hub" on the desktop header bar, or navigate to URL hash "#hub".

### 2. Launch Any Module

Click "Launch Module "" on any card (Ledger, Ingress, MAP Defense, Predict, Design Studio, Procurement).

### 3. Return to Public Website

Click "Public Website" in the upper banner to view the consumer-facing marketing presentation.

TERMINAL REFERENCE

# System Quick Reference & Terminal Commands

*Operational cheat sheet for local development, builds, testing, and production deployment.*

## Command Line Operations

**`npm run dev`**
Starts development server with live Vite middleware on port 3000.

**`npm run build`**
Builds production frontend bundle and standalone Express server.

**`npm start`**
Starts production Node.js service using bundled dist/server.cjs.

**`npm test`**
Executes all 24 Vitest unit tests verifying double-entry posting and gateway.

**`npm run lint`**
Runs tsc --noEmit to validate strict TypeScript type compliance.

> **Support & Compliance Verification**
> All double-entry ledger calculations and canonical schema transformations are certified for GAAP/IFRS audit trails. For custom module integration or API webhook extensions, consult the Universal Gateway definitions in src/gateway.ts.